

phase_1_project

Aviation Accident Analysis

Overview

This project provides a data-driven analysis of aviation accidents to help stakeholders determine which aircraft are low risk and viable investment to purchase for commercial and private enterprises diversifying the companies portfolio expanding to the aviation industry.

Business Understanding

- Frequency occurrence of accidents
- Key factors affecting accident rates
- Level of aircraft damage in relevance to Purpose.of.flight
- Serverity of accident(fatality and survival)
- Least involved make in accidents

Data understanding

Analysis sourced from [Aviation_Data.csv] dataset provided by the National Transportation Saftey Board containing aviation accidents data from 1962-2023 in the United States and international waters.

Key data fields

- Accident.Number
- Event.Date
- Amateur.Built
- Weather.Condition
- Aircraft.damage
- Purpose.of.flight
- Total.Fatal.Injuries
- Total.Serious.Injuries
- Total.Minor.Injuries
- Make

Analysis

✓ Data cleaning

Setup

```
# Importing necessary liabraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# load the csv file into a dataframe df
df = pd.read_csv('Dataset/Aviation_Data.csv', index_col=0, low_memory=
df.head()
```

Event.Id	Investigation.Type	Accident.Number	Event.Date	Location
20001218X45444	Accident	SEA87LA080	1948-10-24	Mt CRE
20001218X45447	Accident	LAX94LA336	1962-07-19	BRIDGE
20061025X01555	Accident	NYC07LA005	1974-08-30	Saltvi
20001218X45448	Accident	LAX96LA321	1977-06-19	EUREK
20041105X01764	Accident	CHI79FA064	1979-08-02	Canto

5 rows × 30 columns

- Handling missing data

```
df.isna().sum().sort_values(ascending=True)
```

Investigation.Type	0
Accident.Number	1459
Event.Date	1459
Location	1511
Make	1522
Model	1551
Amateur.Built	1561

```

Country          1685
Injury.Severity  2459
Registration.Number  2841
Aircraft.damage  4653
Weather.Condition  5951
Total.Uninjured  7371
Number.of.Engines  7543
Purpose.of.flight  7651
Report.Status    7843
Engine.Type      8555
Total.Fatal.Injuries  12860
Total.Minor.Injuries  13392
Total.Serious.Injuries  13969
Publication.Date  16689
Broad.phase.of.flight  28624
Airport.Name     37644
Airport.Code     40216
Latitude         55966
Longitude        55975
Aircraft.Category  58061
FAR.Description  58325
Air.carrier      73700
Schedule         77766
dtype: int64

```

```

# drop misiiing data in the subsets
df=df.dropna(subset=['Accident.Number', 'Event.Date', 'Make', 'Amateur.B
df.head()

```

Event.Id	Investigation.Type	Accident.Number	Event.Date	Location
20001218X45444	Accident	SEA87LA080	1948-10-24	Mt CRE
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5 rows x 30 columns

- Removing duplicated entries

The column Accident.Number is best to identify and remove any duplicated values in the dataset considering its distinct uniqueness.

```
df['Accident.Number'].duplicated().sum()
```

```
np.int64(26)
```

```
df.drop_duplicates(subset='Accident.Number', inplace=True)
```

- groupby Accident.Number by 5years in Event.Date

Grouping the filtered dataset into groups of 5 years to narrow down the large dataset and provide clarity for easy evaluation.

```
# Event.Date is in datetime format
df['Event.Date'] = pd.to_datetime(df['Event.Date'])

#extract 'year' in datetime
df['year'] = df['Event.Date'].dt.year

#group every 5years from 1962-2023
bins=np.arange(1962, 2028, 5)
labels = [f'{i}-{i+4}' for i in bins[:-1]]
df['year_group'] = pd.cut(df['year'], bins=bins, labels=labels, right=False)
df=df.dropna(subset='year_group')
```

Creating a column in the dataset to display counts of each row of the Accident.Number as per every year_group for easier comprehension of the dataset

```
df['number_of_accidents']=df.groupby('year_group',observed=True)['Accident.Number'].count()
df.head()
```

Event.Id	Investigation.Type	Accident.Number	Event.Date	Location
20001218X45447	Accident	LAX94LA336	1962-07-19	BRIDGE
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20001218X45448	Accident	LAX96LA321	1977-06-19	EUREK
20041105X01764	Accident	CHI79FA064	1979-08-02	Canto
20170710X52551	Accident	NYC79AA106	1979-09-17	BOSTO

5 rows × 5 columns

Modelling

Exploratory Data Analysis

In this section , use of Numpy and pandas will be essential to answer the stakeholders questions under business understanding. Matplotlib and pivot tables are used for visualisation.

- Frequency of accidents per year_group

```
frequency_of_accidents=df.groupby('year_group',observed=True)['Accident']  
frequency_of_accidents
```

	year_group	Accidentcount
0	1962-1966	1
1	1972-1976	1
2	1977-1981	4
3	1982-1986	16575
4	1987-1991	13081
5	1992-1996	11421
6	1997-2001	10845
7	2002-2006	9863
8	2007-2011	9285
9	2012-2016	8162
10	2017-2021	7880
11	2022-2026	1581

- Visualisation of the frequency of accidents as per every year_group.
Considering the data from the frequency of accidents is not recognised as a dataframe by the library. To plot extract the frequencies as data for dataframes.

```

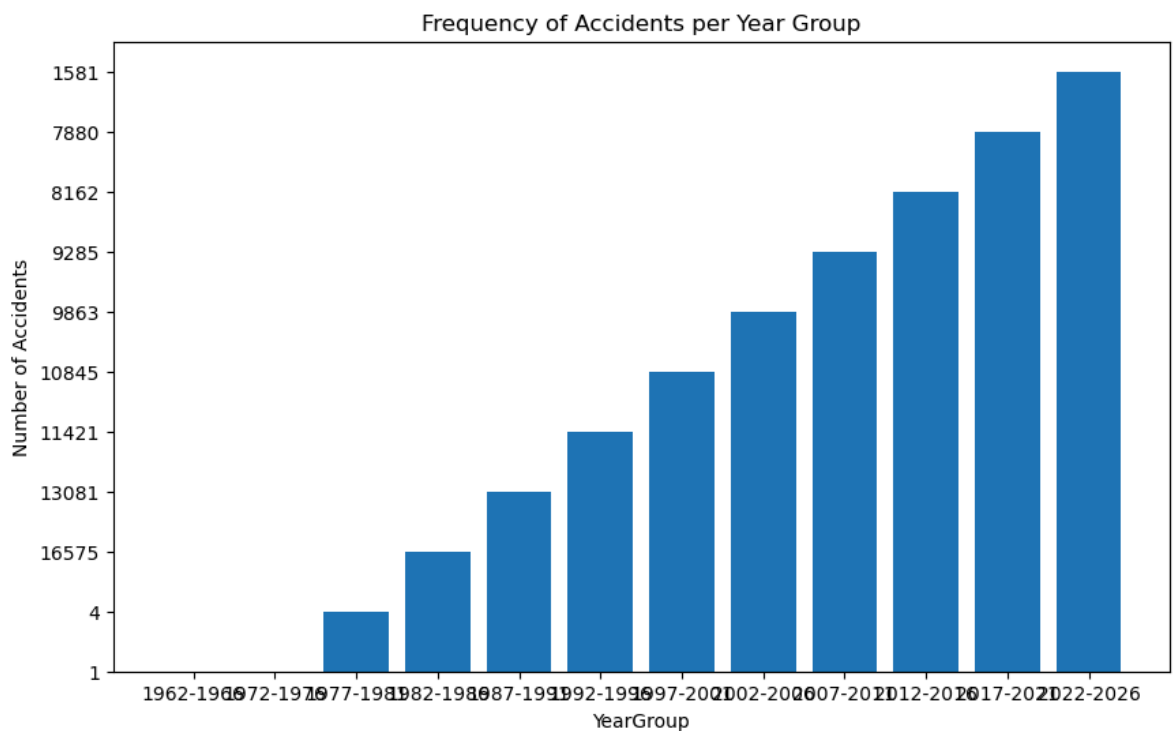
year_groups=year_group=['1962-1966','1972-1976','1977-1981','1982-1986','1987-1991','1992-1996','1997-2001','2002-2006','2007-2011','2012-2016','2017-2021','2022-2026']
Accident_counts=['1','1','4','16575','13081','11421','10845','9863','9285','8162','7880','1581']
data = {'year_group': year_groups, 'Accidentcount': Accident_counts}
frequency_of_accidents = pd.DataFrame(data)

#plots
fig, ax = plt.subplots(figsize=(10, 6))

ax.bar(frequency_of_accidents['year_group'],frequency_of_accidents['Accidentcount'])

# axis
ax.set_title('Frequency of Accidents per Year Group')
ax.set_xlabel('YearGroup')
ax.set_ylabel('Number of Accidents')
plt.show()

```



- Factors affecting accident rate

Filtered Amateur.Built has No to represent Aircrafts not bult for the first time. This information is crucial for finding the number of accidents on every weather condition.

```
plane = df[df['Amateur.Built'] == 'No']
```

```
weather_condition = plane['Weather.Condition'].dropna().str.upper().value_counts()
weather_condition
```

```
Weather.Condition
VMC      69061
IMC      5830
UNK       1043
Name: count, dtype: int64
```

- Level of aircraft damage

The stakeholders are looking to invest in purchase of commercial and private eneterprises planes .This is best represented by

- 'Personal'
- 'Business'
- 'Executive'

in Purpose.of.flight Create a pivot_table to display number of aircrafts to be invested in damage levels.

```
x=['Personal','Business','Executive/corporate']
flight_purpose=df[df['Purpose.of.flight'].isin(x)]

craft_damage=flight_purpose.pivot_table(index='Aircraft.damage', columns='Purpose.of.flight')
print(craft_damage)
```

Purpose.of.flight	Business	Executive/corporate	Personal
Aircraft.damage			
Destroyed	1184	158	10543
Minor	106	32	493
Substantial	2628	330	37989
Unknown	3	2	32

- Serverty of accidents

```
df['total_injuries']=df[['Total.Fatal.Injuries','Total.Minor.Injuries']]
df.head()
```

Event.Id	Investigation.Type	Accident.Number	Event.Date	Location
20001218X45447	Accident	LAX94LA336	1962-07-19	BRIDGE
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5 rows x 34 columns

- Least accident involved make

Least make and model of aircraft determined by the number of accidents incurred and total number of injuries on the accident.

```

# 1. Filter for x purposes and minor aircraft damage
x = ['Personal', 'Business', 'Executive']
df_filtered = df[
    df['Purpose.of.flight'].isin(x) &
    (df['Aircraft.damage'].str.lower() == 'minor')
]

# 2. Group by Make and Model
accident_summary = (
    df_filtered.groupby(['Make', 'Model'])
        .agg(
            Accident_Count=('Model', 'size'),
            total_injuries=('total_injuries', 'sum')
        )
        .reset_index()
)

# 3. Filter for accident and injuries
filtered = accident_summary[
    (accident_summary['Accident_Count'] == 1) &
    (accident_summary['total_injuries'] == 0)
]

# 4. Random pick
sample_size = min(6, len(filtered))
random_samples = filtered.sample(n=sample_size, random_state=42).reset_index()

print('Least involved make in accidents')
print(random_samples)

```

	Make	Model	Accident_Count	total_injuries
0	CESSNA	421	1	0
1	New Piper Aircraft, Inc.	PA-46-350P	1	0
2	COLUMBIA	LC41	1	0
3	STARDUSTER II	SA-300	1	0
4	Cessna	337D	1	0
5	Cessna	A150L	1	0

Conclusion